PHYS 104 – PHYSICS II

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Lecture 3

Dispersion

Monochromatic Light:

- Light that has only one specific wavelength, frequency, and pure color.
- Appears as a single, pure color to the human eye.
- **Examples:** Laser beams and sodium vapor lamps.

Polychromatic Light:

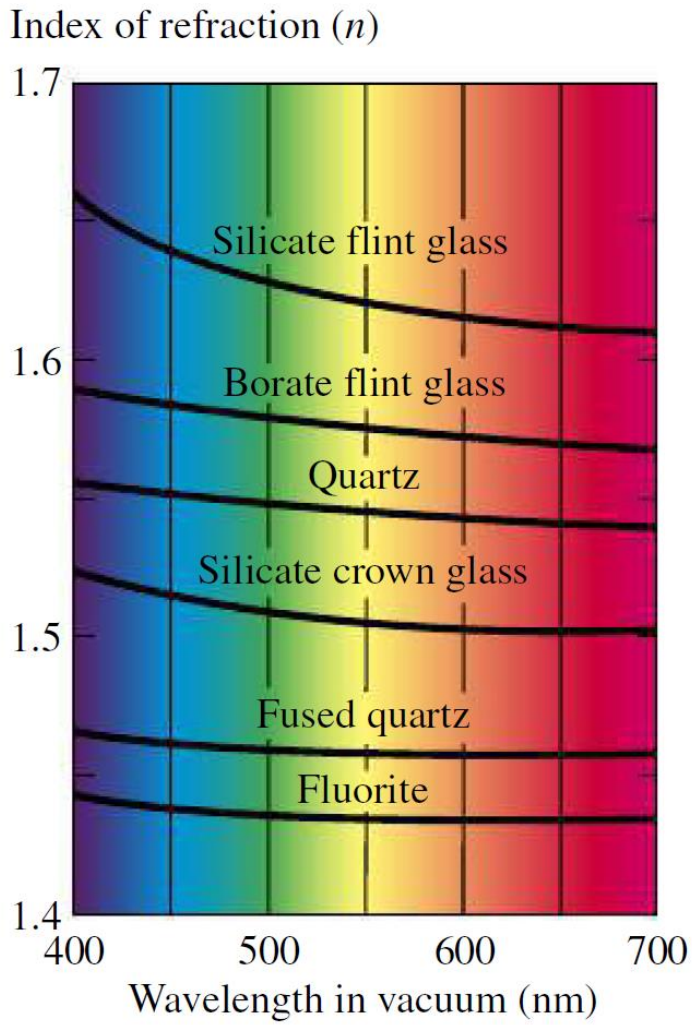
- Light that is composed of many different wavelengths or frequencies combined together.
- Appears as a mixture of colors or as white light containing a full spectrum.
- **Examples:** Sunlight and standard incandescent light bulbs.

Dispersion

- The speed of light in vacuum "c" is the same for all wavelengths, but the speed in a material substance is different for different wavelengths. Therefore the index of refraction of a material depends on wavelength.
- Dispersion is the dependence of the refractive index of a medium (or wave speed) on the wavelength (or frequency) of light,

$$\boldsymbol{n = n(\lambda)}$$

- Because n is a function of wavelength, Snell's law of refraction indicates that light of different wavelengths is refracted at different angles when incident on a material.
- Because $v = c/n(\lambda)$, different colours travel at different speeds in the medium.

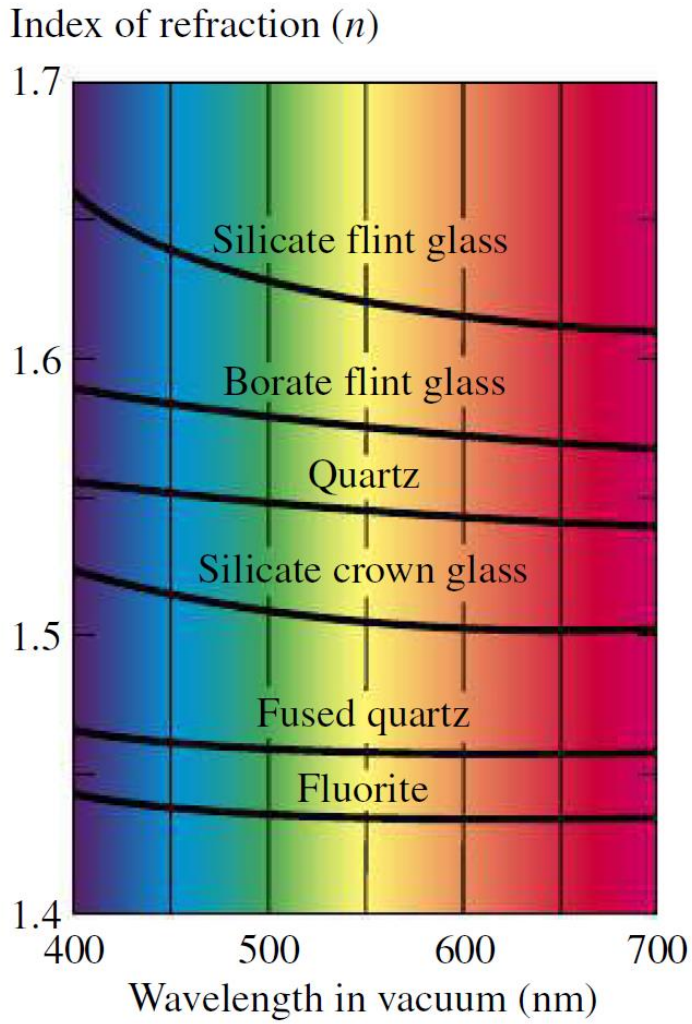


Dispersion

- Over the visible range, n of most transparent materials decreases as λ increases.

$$n_{violet} > n_{blue} > n_{green} > n_{yello} > n_{red}$$

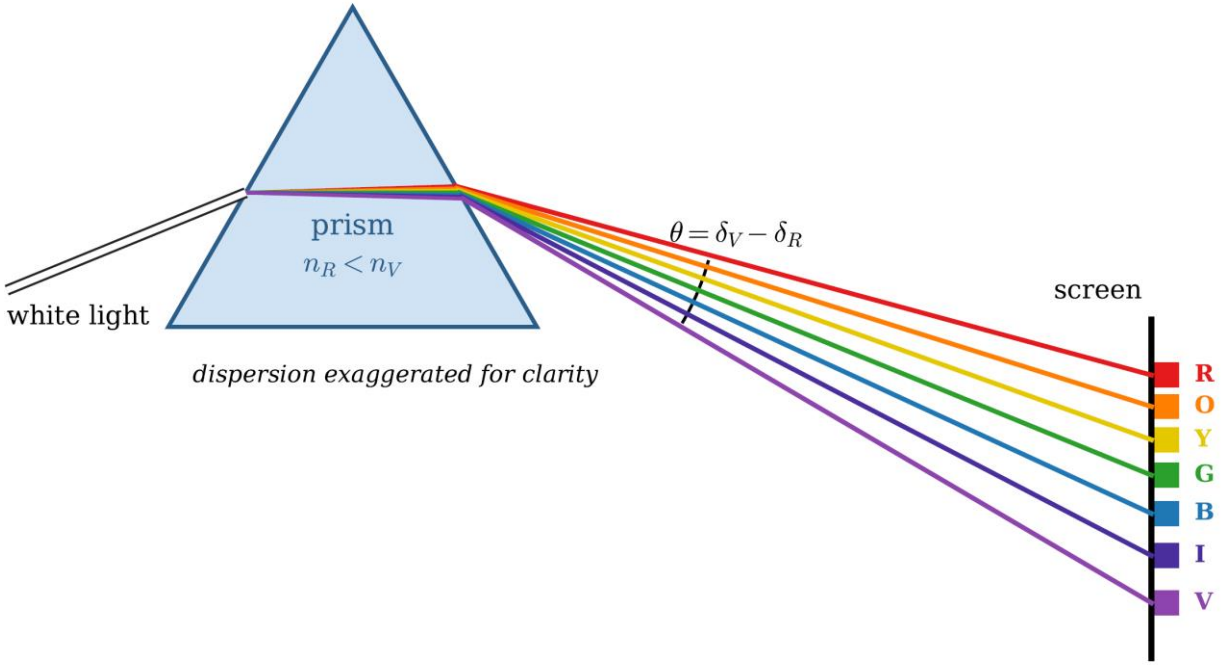
Colour	Vacuum wavelength (nm)
Violet	380–450
Blue	450–495
Green	495–570
Yellow	570–590
Orange	590–620
Red	620–750

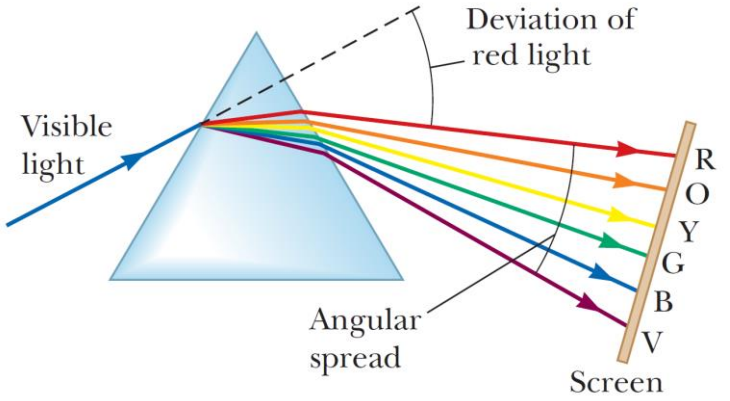


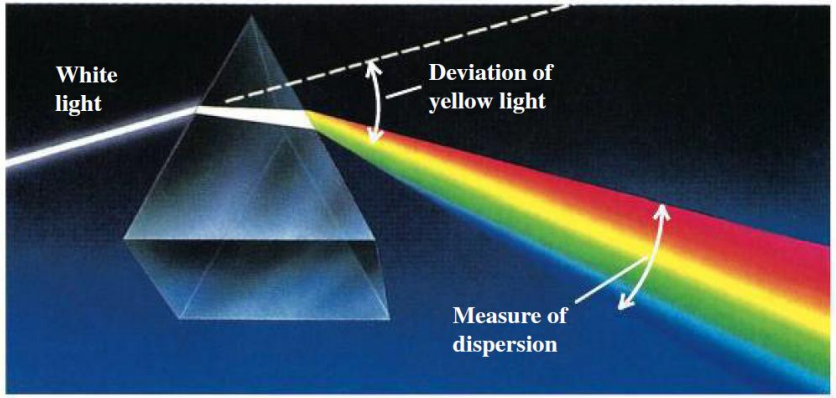
Dispersion

$$\lambda_V < \lambda_R \rightarrow n_V > n_R \rightarrow v_V < v_R \rightarrow \theta_V < \theta_R \rightarrow \delta_V > \delta_R$$

$$\therefore \textit{violet deviates more than red}$$





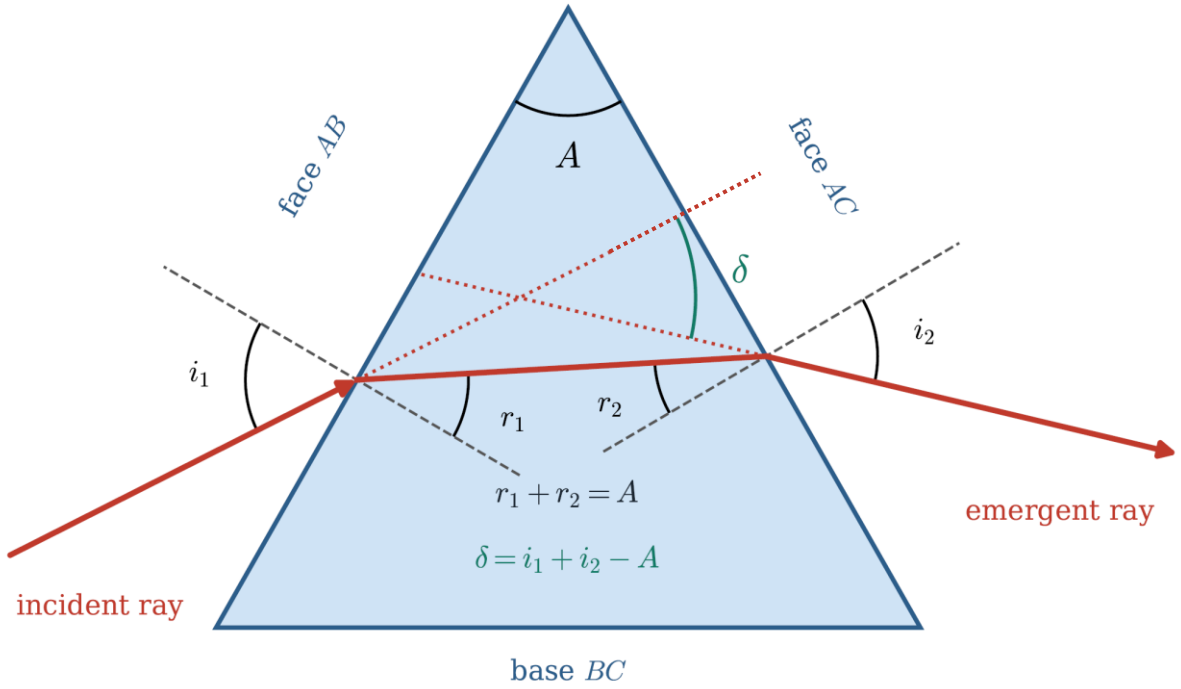


The prism and the deviation angle

- A prism is a transparent medium bounded by two plane refracting faces that meet along an edge.

- **Refracting faces**: the two inclined faces AB and AC.
- **Apex (refracting) angle** A: the angle between the faces.

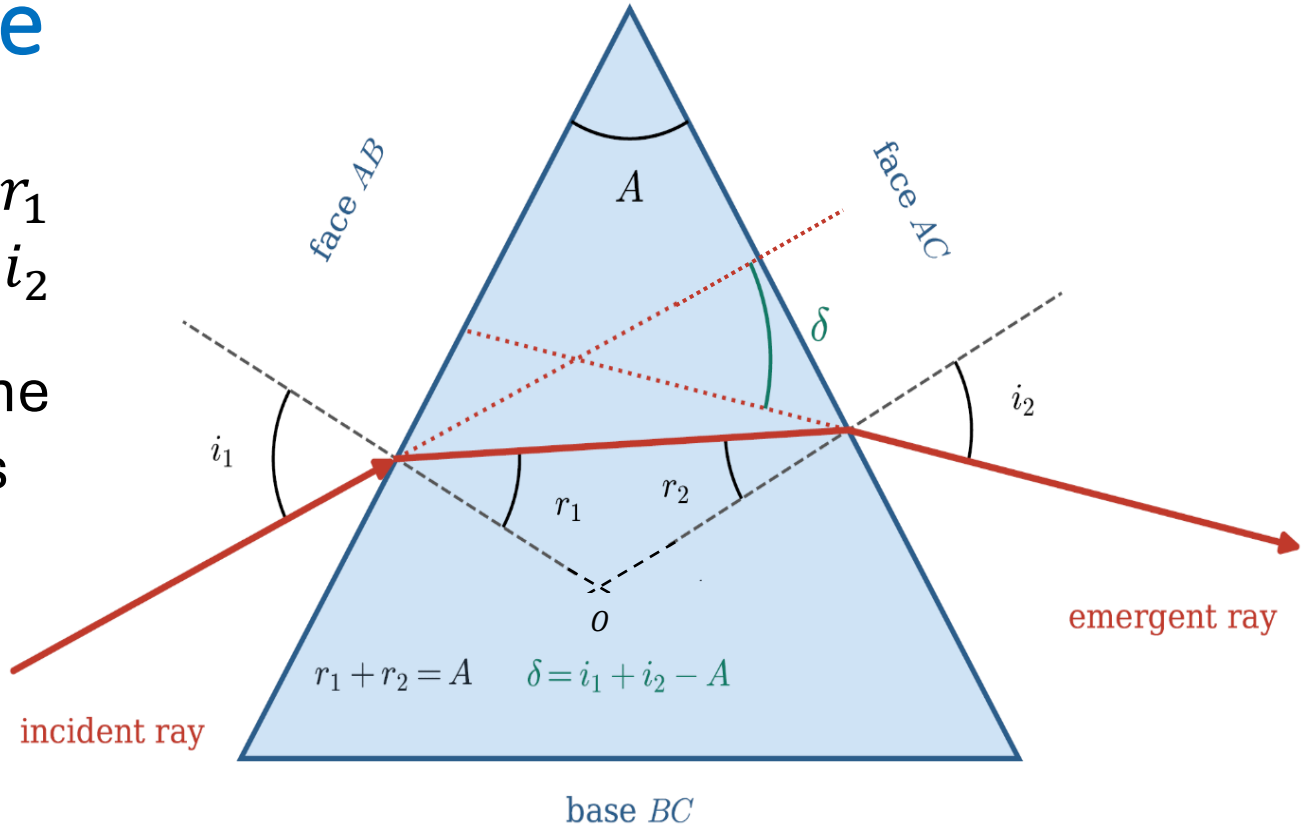
Symbol	Meaning
i_1	angle of incidence at the first face
r_1	angle of refraction at the first face
r_2	angle of incidence at the second face (inside the prism)
i_2	angle of emergence at the second face
A	apex angle
δ	angle of deviation



The prism and the deviation angle

Snell's law at each face:

$$\text{Face } AB: \quad \sin i_1 = n \sin r_1$$
$$\text{Face } AC: \quad n \sin r_2 = \sin i_2$$



- In the quadrilateral formed by the apex, the two feet of the normals and O, two angles are $90°$, so the angle at O is $180° - A$

$$r_1 + r_2 + (180° - A) = 180°$$

$$\boxed{r_1 + r_2 = A}$$

- At the first face the ray is turned by $\delta_1 = (i_1 - r_1)$, at the second by $\delta_2 = (i_2 - r_2)$, in the same sense. The total deviation angle between emergent and incident rays is

$$\delta = \delta_1 + \delta_2 = (i_1 - r_1) + (i_2 - r_2) = i_1 + i_2 - (r_1 + r_2),$$

$$\boxed{\delta = i_1 + i_2 - A}$$

The prism and the deviation angle

Condition for emergence

- If $A \leq \theta_c$, then $r_2 = A - r_1 \leq \theta_c$ and the ray always emerges.
- A ray can emerge only if $A < 2\theta_c$. For $A > 2\theta_c$ no ray can pass, since $r_1 + r_2 = A$ with both $r_1, r_2 \leq \theta_c$ is impossible.
- For $\theta_c < A < 2\theta_c$ there is a *minimum angle of incidence* below which TIR occurs at the second face:

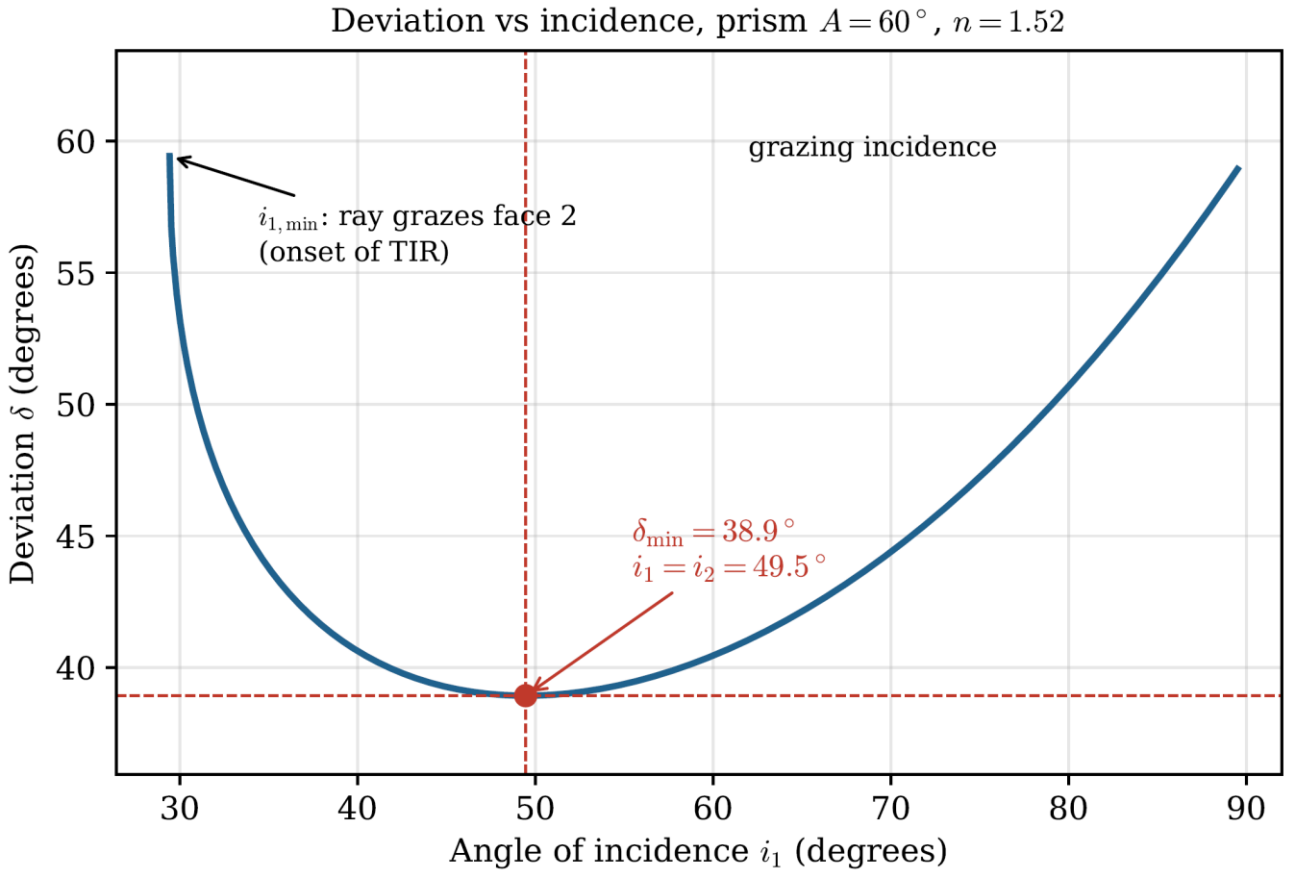
$$\sin i_{1,\min} = n\sin(A - \theta_c)$$

$r_1 = A - r_2 = A - \theta_c$

The prism and the deviation angle

Variation of δ and Minimum Deviation

- As i_1 is increased from its smallest allowed value, δ first *decreases*, reaches a minimum $\delta_{\min}$, then *increases*. The curve is symmetric about the minimum because ray paths are reversible ($i_1 \leftrightarrow i_2$ gives the same δ).



Minimum Deviation Condition

$$\boldsymbol{i_1 = i_2 = i, \qquad r_1 = r_2 = r = \frac{A}{2}}$$

- The ray passes **symmetrically** through the prism.
- Inside an isosceles prism the ray is **parallel to the base**.
- From $\delta_{\min} = 2i - A$: $\quad i = \frac{A+\delta_{\min}}{2}, \quad r = \frac{A}{2}$
- Applying Snell's law at the first face, $n = \frac{\sin i}{\sin r}$:

$$\boldsymbol{n = \frac{\sin\left(\frac{A + \delta_{min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}}$$

This is the standard method for measuring n with a spectrometer. It is accurate because δ is flat near its minimum, so small errors in setting i_1 barely change the measured $\delta_{\min}$.

Dependence of $\delta_{\min}$ on n and A

From $\sin\frac{A+\delta_{\min}}{2} = n\sin\frac{A}{2}$:

- $\delta_{\min}$ increases with n (fixed A);
- $\delta_{\min}$ increases with A (fixed n);
- since $n_V > n_R$, $\delta_{\min}(V) > \delta_{\min}(R)$.

Thin-prism approximation

For a thin prism ($A \lesssim 10°$) with small angles, replace sines by angles: $i_1 = nr_1$, $i_2 = nr_2$. Then

$$\delta = i_1 + i_2 - A = n(r_1 + r_2) - A = nA - A,$$

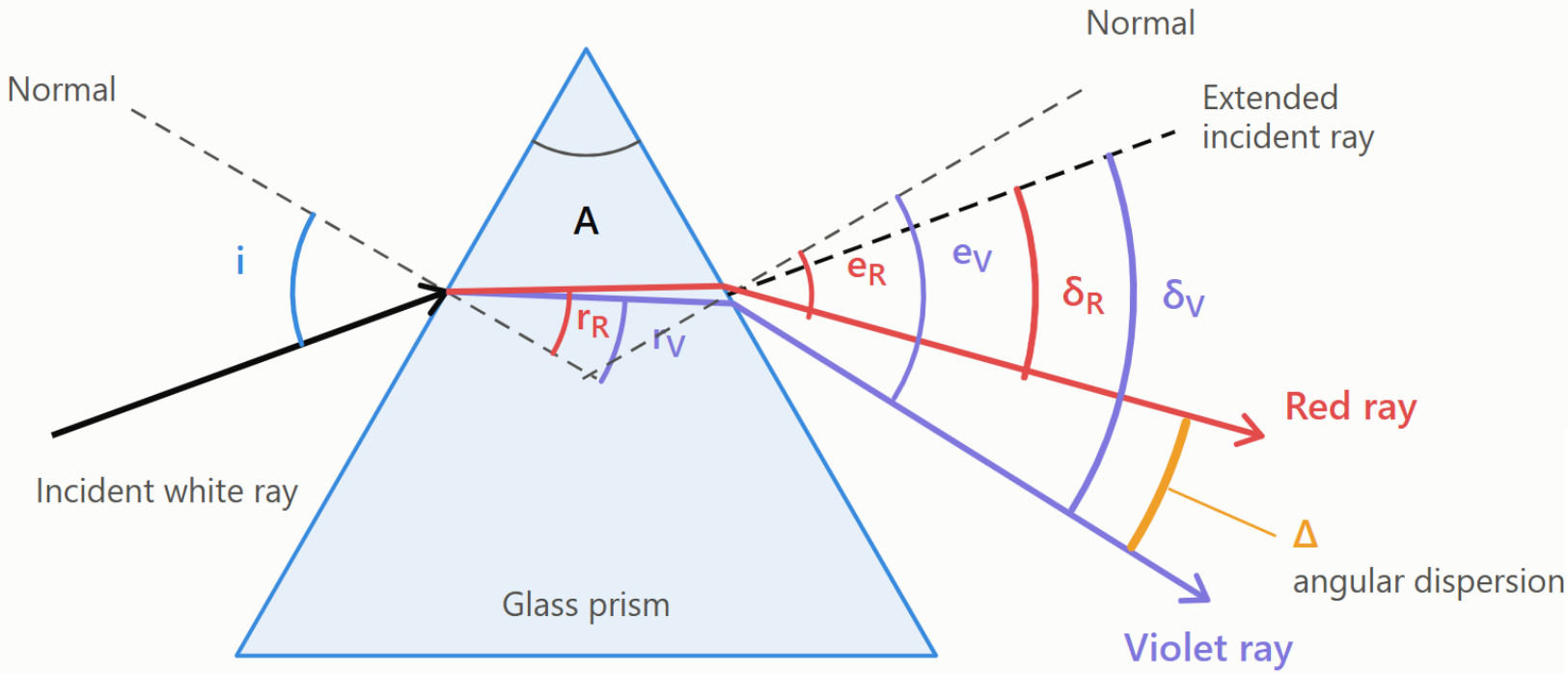
$$\boldsymbol{\delta = (n-1)A}$$

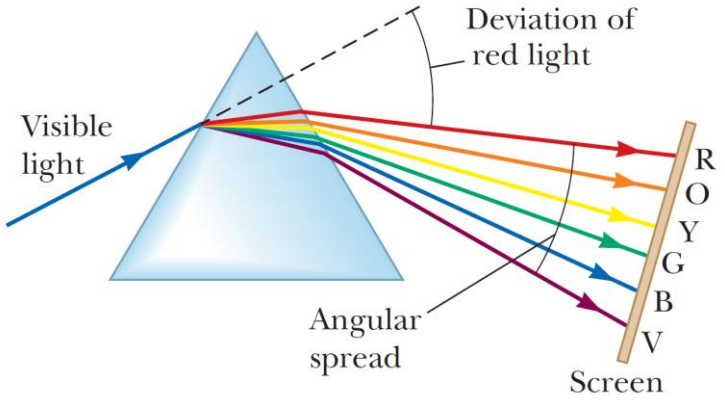
independent of the angle of incidence.

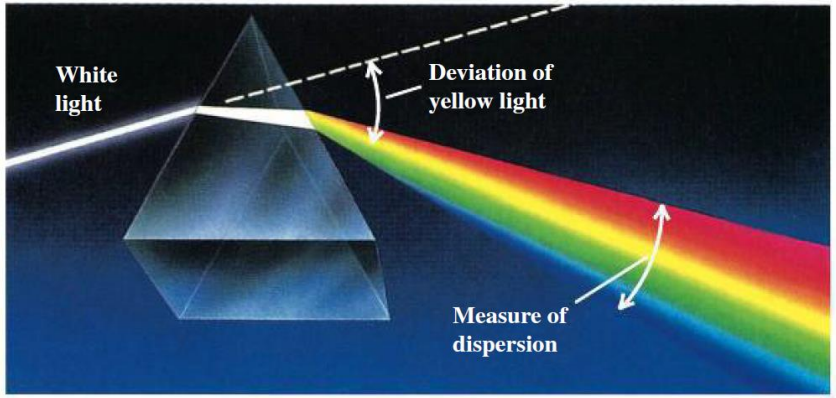
Dispersion by a Prism

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Dispersion by a Prism

Angle of dispersion

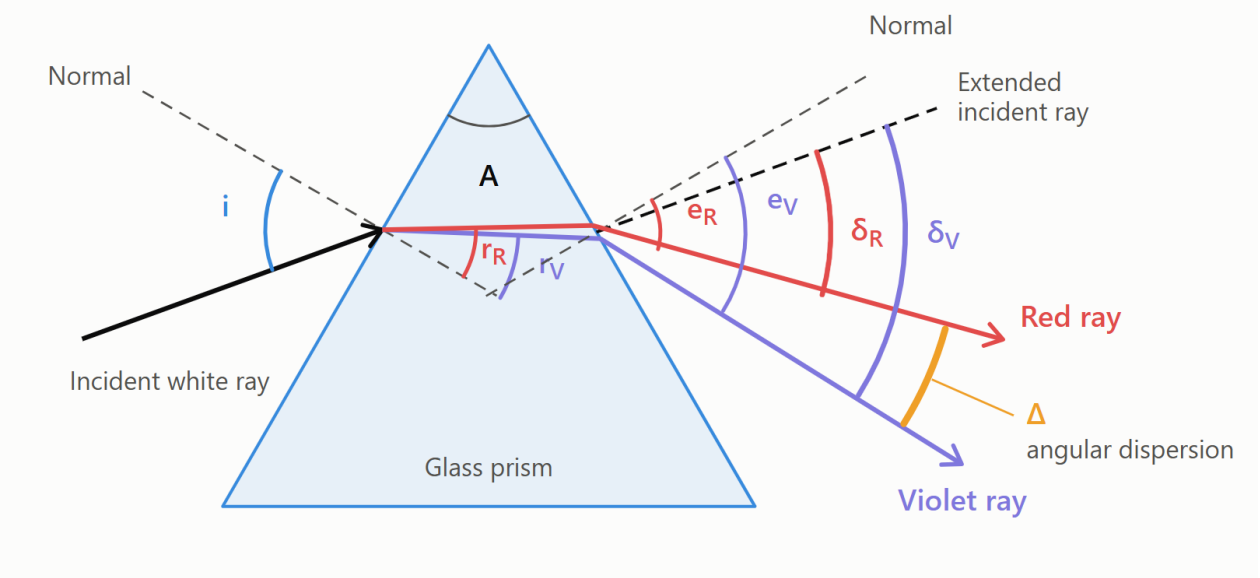
- The angular separation between the emergent rays of two chosen colours, usually violet and red:

$$\boldsymbol{\theta = \delta_V - \delta_R}$$

- For a thin prism, using

$$\theta = (n_V - 1)A - (n_R - 1)A = (n_V - n_R)\,A.$$

It is proportional to A and to $n_V - n_R$, so it is *not* a pure material constant.



Dispersive power and the Abbe number

- To characterize the *material* independently of the prism angle, compare the angular dispersion with the mean deviation.

$$\omega = \frac{\delta_V - \delta_R}{\delta_Y}$$

- For a thin prism, $\delta = (n-1)A$, so

$$\omega = \frac{(n_V - n_R)A}{(n_Y - 1)A} = \frac{n_V - n_R}{n_Y - 1}.$$

- Properties of ω:

1. It is dimensionless.
2. It depends on the material only, not on the prism angle.
3. It is positive for normal dispersion.
4. A larger ω means more colour spread per unit mean bending.

$$V = \frac{1}{\omega} = \frac{n_Y - 1}{n_V - n_R}.$$

High V (crown glass, $V > 50$) means low dispersion; low V (flint glass, $V < 50$) means high dispersion.

Example 1:

A prism of apex angle $60°$ gives $\delta_{\min} = 37°$. Find n.

Solution

$$n = \frac{\sin\frac{A + \delta_{\min}}{2}}{\sin\frac{A}{2}} = \frac{\sin 48.5°}{\sin 30°} = \frac{0.7490}{0.5} \approx \mathbf{1.50}$$

Example 2:

A 60° BK7 prism ($n_R = 1.5143, n_V = 1.5224, n_Y = 1.5170$) is used at minimum deviation. Find δ_R, δ_V, θ, ω and V.

Solution

$$n = \frac{\sin\frac{A+\delta_{\min}}{2}}{\sin\frac{A}{2}} \quad \rightarrow \quad \frac{A+\delta_{\min}}{2} = n\sin 30 = n/2$$

- Red: $\sin\frac{60^\circ+\delta_R}{2} = \frac{n_R}{2} = 0.75715 \Rightarrow \frac{60^\circ+\delta_V}{2} = 49.22^\circ \Rightarrow \delta_R \approx 38.44^\circ$.
- Violet: $\sin\frac{60^\circ+\delta_V}{2} = \frac{n_V}{2} = 0.76120 \Rightarrow \frac{60^\circ+\delta_V}{2} = 49.57^\circ \Rightarrow \delta_V \approx 39.14^\circ$.

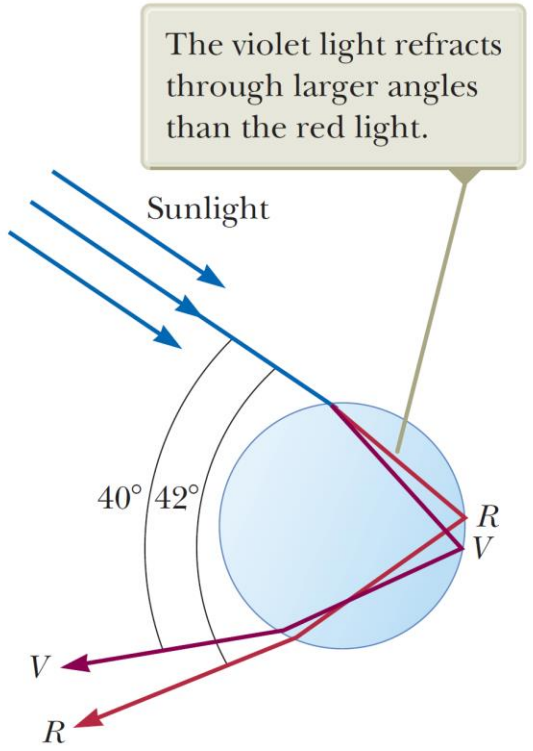
$$\theta = \delta_V - \delta_R \approx \mathbf{0.70^\circ}$$

$$\omega = \frac{n_V - n_R}{n_Y - 1} = \frac{1.5224 - 1.5143}{1.517 - 1} = \mathbf{0.0156}$$

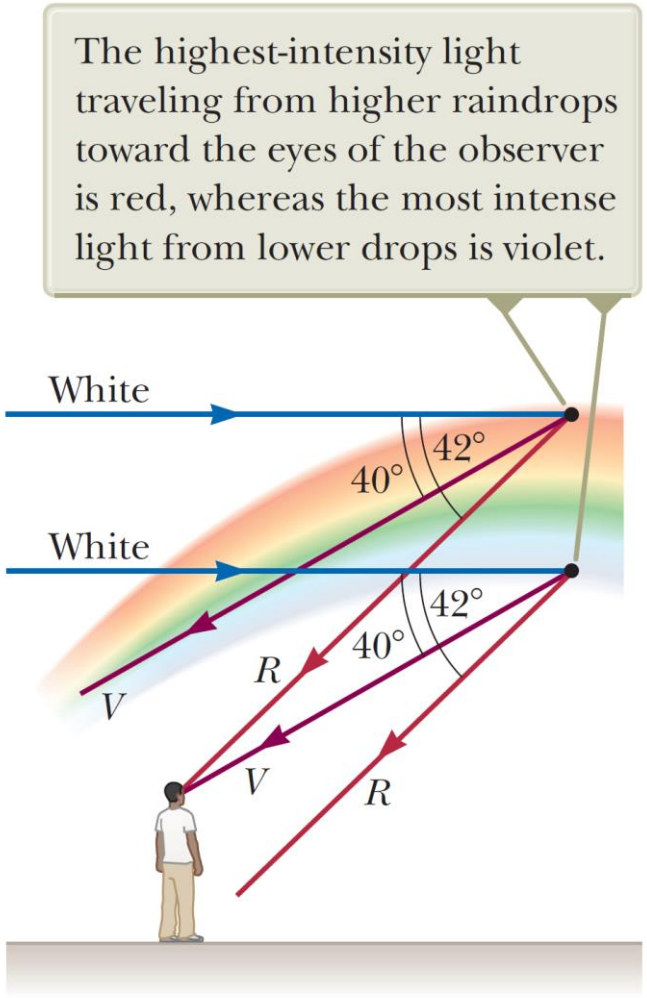
$$V = \frac{1}{\omega} = \frac{1}{0.015} = 64.1 > 50 \quad \rightarrow (\text{low dispersion})$$

Rainbow

- A ray of light passing overhead strikes a drop of water in the Atmosphere and is refracted and reflected as follows: it is first refracted at the front surface of the drop, with the violet light deviating the most and the red light the least.
- At the back surface of the drop, the light is reflected and returns to the front surface, where it again undergoes refraction as it moves from water into air.
- The rays leave the drop so that the angle between the incident white light and the returning violet ray is 40° and the angle between the white light and the returning red ray is 42°. This small angular difference between the returning rays causes us to see the bow.
- Now consider an observer viewing a rainbow, as If a raindrop high in the sky is being observed, the red light returning from the drop can reach the observer because it is deviated the most, but the violet light passes over the observer because it is deviated the least.
- Hence, the observer sees this drop as being red. Similarly, a drop lower in the sky would direct violet light toward the observer and appear to be violet.



Path of sunlight through a spherical raindrop. Light following this path contributes to the visible rainbow.



The formation of a rainbow seen by an observer standing with the Sun behind his back.